

Yiming (Emmett) Peng

 [emmett-peng.github.io](https://github.com/emmett-peng) |  emmett.peng@mail.utoronto.ca |  [Emmett Peng](#) |  [Emmett-Peng](#)

EDUCATION

- **University of Toronto** Sept 2026 (Expected)
Doctor of Philosophy in Biostatistics
Research areas: Bayesian causal inference, spatial data analysis, and machine learning
◦ Advisors: Kuan Liu, Zihang Lu
- **University of Toronto** Sept 2024 – Aug 2026 (Expected)
Master of Science in Biostatistics – Thesis Option; GPA: 4.00/4.00
Thesis: Bayesian Nonparametric Latent Class Framework for Causal Inference in Cognitive Trajectories
◦ Advisors: Kuan Liu, Zihang Lu
- **Beijing Institute of Technology, Zhuhai** Sept 2020 – Jun 2024
Bachelor of Science in Applied Statistics; GPA: 3.96/4.00 (WES)
Thesis: Deep Learning and Stochastic Modeling of Human Activity Patterns through Smartwatch Sensors for Health Interventions (*Honored as Outstanding Thesis*)
◦ Advisor: Han C.W. Hsiao

PUBLICATIONS AND MANUSCRIPTS

R=UNDER REVIEW, S=IN SUBMISSION, W=WORKING PAPERS

- [R] Denier, D., Peng, Y. (Emmett), ..., Ricciuto, A. (2026) **Longitudinal GGT Trajectories Identify Prognostic Phenotypes in Pediatric Primary Sclerosing Cholangitis**. *Under review at JHEP Reports*.
- [S] Peng, Y. (Emmett)*, Truong, V.*, Xie, A., Shi, Y., Hu, P. (2026) **PRECISE: A Framework for the Prediction of Response Using Cell-Type Inference and Single-Cell Embedding in Breast Cancer Patients Receiving Anti-PD-1 Treatment**. *In Submission*. [\[Code\]](#)
- [W] Peng, Y. (Emmett), Lu, Z., Liu, K. (2026) **Bayesian Nonparametric Latent Class Framework for Causal Inference in Cognitive Trajectories**. *Manuscript in preparation*.
- [W] Zhang, J., Shi, Y., Xie, A., Peng, Y. (Emmett), Hu, P. (2026) **Predicting New-Onset Atrial Fibrillation with Competing Risks Using an Uncertainty-Calibrated Interpretable Tabular Transformer: Evidence from the Cardiovascular Imaging Registry of Calgary (CIROC)**. *Manuscript in preparation*.
- [W] Khan, A., Peng, Y. (Emmett), Zhang, I., Chen, H. (2026) **Robust and Interpretable Machine Learning for Atypical Parkinson's Disease Classification Under Distribution Shift**. *Working paper*.
- [W] Jing, Y., Peng, Y. (Emmett), Danguedan, A., Knight, A. (2026) **Examining the Relationship Between Socioeconomic Factors and Mental Health in Childhood-Onset Systemic Lupus Erythematosus Using a Multilevel Analysis**. *Working paper*.

PRESENTATIONS

- [1] Peng, Y. (Emmett), Lu, Z., Liu, K. (2026) **Bayesian Nonparametric Latent Class Framework for Causal Inference in Cognitive Trajectories**. Accepted presentations at the *Statistical Society of Canada (SSC) Annual Meeting, Joint Statistical Meetings (JSM)*, and *ICSA-Canada Chapter Symposium*, 2026. *Forthcoming*.
- [2] Peng, Y. (Emmett), Denier, D., ..., Ricciuto, A. (2026) **Longitudinal GGT Trajectories Identify Prognostic Phenotypes in Pediatric Primary Sclerosing Cholangitis**. Poster presentation, *Canadian Statistics Student Conference (CSSC)*, Hamilton, Ontario, Canada, May 2026. *Forthcoming*.
- [3] Peng, Y. (Emmett), Truong, V., Xie, A., Shi, Y., Hu, P. (2025) **Enhancing Breast Cancer Treatment Response Prediction with Single-Cell RNA Sequencing and Large Language Models**. Oral presentation and poster, *2025 Statistical Society of Canada (SSC) Annual Meeting*, p. 58. May 25–28, Saskatoon, Saskatchewan, Canada. [\[Slides\]](#) [\[Poster\]](#)
- [4] Zhang, J., Shi, Y., Xie, A., Peng, Y. (Emmett), Hu, P. (2025) **Uncertainty-Calibrated Interpretable Tabular Transformer Model for Atrial Fibrillation Prediction with Competing Risk**. Poster, *2025 Statistical Society of Canada (SSC) Annual Meeting*, p. 42. May 25–28, Saskatoon, Saskatchewan, Canada. [\[Poster\]](#)
- [5] Peng, Y. (Emmett), Lu, Z., Liu, K. (2025) **Bayesian Latent Class Causal Inference with Nonparametric Covariate Modeling for Endpoint Cognitive Outcomes in Aging Populations**. Invited poster presentation, *Data Science Institute Causal Inference Workshop*, Toronto, Ontario, Canada, October 22, 2025. [\[Poster\]](#)

- **Knight Lab, The Hospital for Sick Children** Jun 2025 - Present
Clinical Research Project Assistant, supervised by Dr. Andrea Knight
◦ Implement Bayesian statistical methods in R to study socioeconomic and mental health outcomes in childhood-onset systemic lupus erythematosus (cSLE).
◦ Lead statistical analyses, including data pre-processing, imputation, and modeling, collaborate on study design, and provide statistical consulting while contributing to manuscript preparation.
- **The Hospital for Sick Children** Jun 2025 - Present
Research Volunteer, supervised by Dr. Amanda Ricciuto and Dr. Anne Griffiths
◦ Implement latent class mixed models in R to analyze longitudinal serum biomarker trajectories in primary sclerosing cholangitis (PSC).
◦ Integrate latent class membership into survival analysis to assess prediction of transplant outcomes, and collaborate with clinicians and biostatisticians on interpretation and manuscript preparation.
- **Hu Lab, Dalla Lana School of Public Health, University of Toronto** Oct 2024 - Jun 2025
Data Science Research Student, supervised by Dr. Pingzhao Hu
◦ Developed an integrated multi-stage pipeline in Python combining single-cell RNA-seq data with embeddings from large language models (LLMs) and foundation models to identify cell type-specific biomarkers for breast cancer treatment response.
◦ Implemented supervised classification using cell-type-specific biomarkers and cluster-specific bulk gene-expression profiles to predict treatment response, achieving improved predictive performance over baseline.
- **The University of Hong Kong** Jan 2025 - Apr 2025
Research Volunteer, supervised by Dr. Liwu Zheng
◦ Collaborate with clinicians on biostatistical research projects.
◦ Provide statistical consulting for clinical studies.

<i>Amounts in Canadian dollars unless otherwise specified</i>	
• ICSA-Canada Chapter Symposium Student/Postdoc Travel Scholarship Award - \$600 <i>ICSA-Canada Chapter Symposium</i>	2026
• Winner – Case Studies in Data Analysis Competition, SSC 2025 - \$2,500 per team <i>Statistical Society of Canada (SSC)</i>	2025
• CSSC 2025 Travel Award - \$150 <i>Canadian Statistics Student Conference (CSSC), sponsored by CANSSI</i>	2025
• SGS Conference Grant - \$450 <i>University of Toronto</i>	2025
• BITZH President’s Scholarship - \$2,000 <i>Beijing Institute of Technology, Zhuhai</i>	2023
• College Dean’s Honor Scholarship - \$7,000 <i>Beijing Institute of Technology, Zhuhai</i>	2021 - 2024
• BITZH Freshman Entrance Scholarship and Continuing Scholarship - \$18,000 <i>Beijing Institute of Technology, Zhuhai</i>	2020 - 2024
• BITZH Outstanding Student Scholarship <i>Beijing Institute of Technology, Zhuhai</i>	2020 - 2024

- **Programming Languages:** R, Python, MATLAB, C++
- **Tools & Technologies:** Git, Linux, HPC (Compute Canada, Slurm), LaTeX, RMarkdown, Jupyter Notebook
- **Statistical Methods:** Causal Inference, Latent Class Analysis, Survival Analysis, Longitudinal Data Analysis
- **Machine Learning Models:** Transformers, RNNs, Regression (GLM, Penalization), CNNs, Neural Networks
- **Languages:** Chinese (native), English (professional working proficiency)

• ASA DataFest at UofT <i>Judge and Mentor</i> • Biostatistics Union of Graduate Students (BUGS) <i>PHSA MSc Representative and Seminar Committee member</i> • Health Data Working Group <i>Student Member</i> • Public Health Students' Association, University of Toronto <i>Biostatistics MSc Representative</i> • Student Union, College of Global Talent, Beijing Institute of Technology, Zhuhai <i>Officer</i>	May 2026 Sep 2024 - Present Sep 2024 - Present Sep 2025 - Present Sep 2020 – Jun 2022
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